

- **CLR** → Responsible Full Responsibility about the APP From Running to end.

① → **Class loader** → load class from hardisk into memory (when you start use it) → when need it

② → **MSIL To Native Compiler (JIT)**
(Just In time) → Responsible for converting
IL To **Native**

③ → **(Garbage Collector)** → one of the Problem at C/C++
memory management // .Net Framework (the CLR is the Responsible for memory management).

the Component at the CLR is Responsible for memory management is **(Garbage Collector)** → keep track on the memory that you allocate (it). when the Garbage Collector detect that you don't use this memory (this part of memory) it automatically deallocate it.

(How?) at C++ Pointer is **(is)** on address when the function killed the pointer also killed and the space at memory.

→ There is something called (Gc manager)

Two Types of writing Code → ① managed Code
→ ② un managed Code

① managed Code → It has the Responsible for its execution is CLR

② un managed Code → CLR Not Responsible for this Code
→ Like at C# there is a pointers at C# .
But we don't need to use it.

→ Can Book memory and Refer to it with pointer at C#
→ Then the Unmanaged Code → the CLR Not Responsible
→ Garbage Collector not tracked it.

You are the Responsible for deallocate it.

→ why Like 1% still Use un managed Code ??

You are able to integrate C# Code with C++ Code with COM marshaler. (at CLP)

C++ IS Library ← سائل
وإن كان ال app يأتي بقدر من جهاز ال سائل
Integration مع ال Libraries بأجهزة

another Important thing is (DeBug Engine)

→ software Helps Us To DeBug app while Running
Trace app (watch Items and Variables)

(Security Engine) → some of Rules executing at the
Run Time and this Engine is the Responsible

(Type checker) → C# is a strongly Typed language

→ at the Compile Time and the Runtime, we should
Be ~~the~~ sure that → the variable ~~at~~ assigned int
Value the same data type

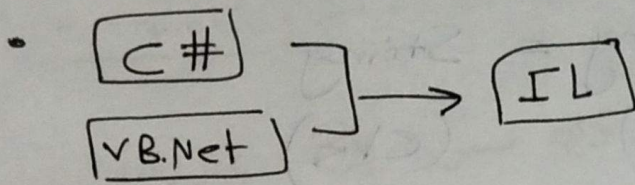
→ (exception manager) → there are Runtime errors
happens, the Responsible
for handling this exceptions is the exception manager.

• at C, C++ → Runtime error occurs the APP
Crashes.

exception manager handles that

→ (Thread support) → the one Process can ~~be~~ extract
child Process from it called Threads (many
(16)

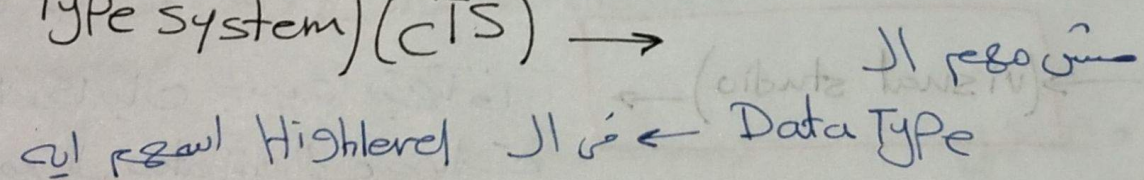
→ all of these that we are talking about should contain [Base class Library support]



- Syntax is different Between C# and VB.Net
Not just the syntax But also the data types

- C# → (int) } when converting to (IL)
VB.Net → (Integer) } there was a difference code
But also the same concept !!

- that's why (CTS) → has been out.

→ (Common Type System) (CTS) → 
High Level → IL → Data Type

• IL → Data Type

- when type in like C# language Two names

~~For data~~ For data types

- ① C# name
- ② CLR name

when you are typing C#
C# or CLR name.

You can type

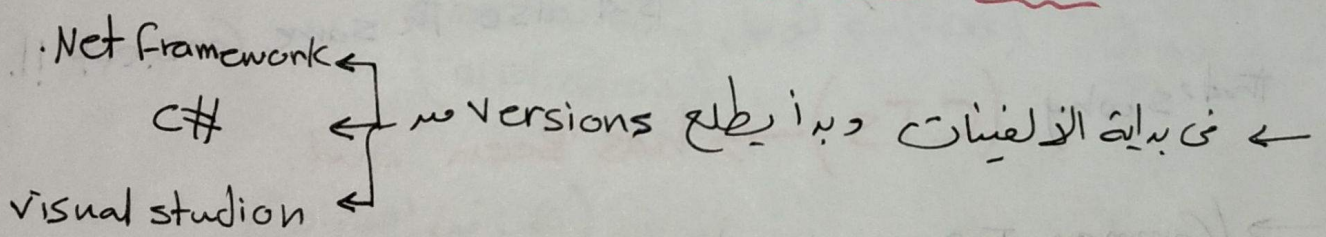
→ Int = Int 32 → (CLR)
(C#) (~~C#~~)

→ Long = Int 64
↓
C#

string = String
(C#) (CLR)

- That's we have finished the Components of
DCLR ② .Net Framework

(Revision of History of the .Net Framework)



→ (visual studio) →

علاوة فيه اوكيان جديدة بتضاف
علاوة يتماشى مع ال updates التي حصلت
في ال .Net framework و C# كلغة

[2002 / 2003 / 2005 / 2008 / 2010 / 2012 / 2015 / 2019 / 2022]

→ .Net Framework → 1.0 / 1.1 / 2.0 / 3.0 / 3.5 / 4.0 / 4.5
4.6 / 4.7 / 4.8 / —

← يا اما التغيير في مستوى ال نظام . ☐ ☐ ☐

لو التغير على مستوى القابلية الحال في الرقم 1, 2, 3 -

وهو معنا انه فيه major updates وصلت في ال Framework

وهو Update على مستوى .Net . كل من C# و CLR + BCL

(.NetFramework)

الموضوع فضل كده (4.8) و كان ظهر في الساعة

بالتوازي مع ال updates بتاع ال .NetFramework

حاجة اسماها .Net Core ← 2, 3, 3.0, -

الهدف منه ال (core) ← توصل (CrossPlatform application)

← من سنة ما ← Microsoft قالت (هتوقف اصدارات .Net Framework)

وهكذا Investment على مستوى ال Core ، و هذا رايي

ال Runtime هتبقى اسماها (.Net) بين من ← Core Framework

versions

من اللى في بيها يظهر

.Net 5, 6, 7, 8, 9 -

بالتوازي مع الـ C# يحصل ده ، كانه بيـه updates + مستوى

الـ C# كلغة يعني . آخرهم (12) C# بيايه

← هفتنح الـ (IDE) دلوقتي (new Project)

/ windows / Console

→ Console APP → .Net

→ Console APP (.Net Framework) → ~~.Net~~

*

← هفتنح الـ Console APP بس

(solution) → the Place that I can put more than one Project inside it.

(long term support) → generally (3) years

→ ☐ Don't use top level statements

check ^{منه} Now.

د (11: 39)

→ solution explorer :

my solution and all Projects I have made If made more than one Project.

→ (Program.cs) → main Function → the Entry Point

C# Case sensitive (Main → ~~is~~ UPPER Case)

→ (C++) → Not fully object oriented but it violates
Somethings / including [stand alone Function]

← عنى Function مكتوبه على الـ class

← لى الكلام ده مش موجود فى الـ C# بل فى الـ Function
تكتب جوا class .

why main static ??

→ when the Application is Running, the Invoker of the OS
(CLR) → needs to executes (main) . If it's not static
→ needs creates object from the class that it's inside
to call it → that's why main should be
static and called automatically / generally it's
Public.


```
Console.WriteLine("Helloworld");
```

→ Function name → (write line)

Public static function

→ Inside Console class, that's why we are calling it with its class name.

→ Run & Debug + Run & (F5) & (Ctrl+F5) → not hit at Break Point

→ Build → (F6)

→ ① where is the (Console class)?

C# → You can create classes at the app, they can put them at (namespace)

(namespace) is ~~referred~~ classes as groups ←
→ logical separation of classes.

on the Hard disk I don't have → class
→ namespace

I have (file) → Can write in it some classes
(or) Put your classes on
more than one file

(22)

→ ال Classes دي سواء موجودة في File واحد
والا أكثر من File .

ينفع (logically) خبهم (Related) الى namespace واحد

* تقسيم logical لـ Classes بتاعتها

← على التنظيم ← منعا للتكرار مثلا

← حبيب لو شغال في app
Book ← تشغل مع كتب
Book ← حجز كتب

→ مينفعش يكون معايا اثنين Classes على نفس ال app كده
نفس الاسم ← له خلى كل واحد منهم في namespace مختلف

(XML) → qualified name of the tag [namespace, tag name]

→ each class is Related to namespace.

name space Day ①

(internal class Program) ← our app

that's mean the class Program is in

namespace Galled (Day one)

(23)

معنی کہ علانیہ اوپر لاء Class Program میں اس صنف 11

← لکھتے Day one program (name space.classname)

(من الآخر) ← البرمجیت صیغہ
exe کتبوں علی مجموعہ میں
DLL
classes
namespaces

By default Putted

at the Day one

(namespace)

لو عاوز اعلیٰ class جہ یہ ←
employee

← سبب انا فلا ہیر ال namespace لای اسم مثلاً (HR)

صوف لاء Class Program و عاوز اوپر ل employee

لو کتب Employee میں میں لکھہ استخدا

ہم اقول HR.Employee

تقسیم لائنہ الکوڈ لایکس لاء و عدداں Classes یکسر

← بندہ خلی کل سٹیو Related Classes
to Business specific inside

(24) specific namespace

→ Project Can Be exe (or) DLL

→ DLL doesn't contain main??

لا تسمى هير

→ Classes / name spaces → No Run

when Build the DLL → will Be executed file DLL

namespaces ال namespace الواحد يقع يحتوي جود
↳ inside it

→ That's for the Code that we are writing
what about the Code we are using?

Console → class Ready To Use // from namespace called (system).

when Using class you should → ① system. Console. writeline
مع كونه كل ما احب ان افعل مع Console class فمفتركتي
قبله system.

→ ② Before name spaces up the codes
→ using (system) → namespace.

→ If I don't type system at first?

at .NetFramework (old) you should type system.
But .Net C# → there is something called global using

→ Physically the namespaces should be at DLL

Logically → namespace at the app.

→ DLL that includes all namespaces —

→ (System)

من كل اسم ال DLL يكون نفس اسم

* ال namespace ال DLL الاله يقع يكون جواه اكثر من namespace

* visual studio ← هو الي مكان ال DLL للبروجيكت

Dependencies → Frameworks.

• If I need to add new DLL → add Project Reference

Browse References.

1) (DataTypes) →

`int x;` ; `int 32 x;`

value Type

Reference Type

→ there are 3 main ways to create new DataType.

1) Class 2) struct 3) enum

any DataType with → struct
→ enum } → Value Type

Class → Reference Type

→ رقم عشري → 1) Float 2) Double 3) Decimal

مستوى دقة Precision هو الفرق بين الثلاث

كام رقم بعد العلامة العشرية

(money) → decimal

(scientific) → Float, double

من فكرة حجم دقة كل واحد

الرقم يتصل اذايه.

1) Precision	(Decimal) >> double/Float
رقم عشري	double >> Decimal

$$\rightarrow (BCL) \rightarrow (IL)$$

→ DataTypes are Related To CLR NOT To The language.

→ ① int x;

$$\text{int } y = x;$$

عام عادى في البداية (X) لتفسير garbage و خلاص.

* دلوقتى لا ما هذا ليجو anymor

↳ Use ~~an~~ unsigned variable (✓).

assign Before Use

① ← جس ال local variables

→ each data type has Default value.

- int → 0
- Boolean → False

*If I want to initialize variable and want to

~~Point~~ ~~the~~ assign it the default value.

① int x = 0;

② int x = default

* له حل في بعض الاوقات فنظر لستخدام default و

• Templates ١٥ و ١٦ ; Generics ١١ و ١٢

← ال Compiler عنه اللغة، على التفرقة على ال literants

← الفصح المطلقة

int x = 123;

$$\dot{F}_{\text{load}}^* = 5.8 \text{ i} + \text{t}$$

1) Float $y = 5.8;$

The Compiler ~~also~~ always consider this a double

ملاحظة
بالنسبة

There is no implicit conversion at the Compiler from double to float.

(sol) Float $y = 5.8$ (FF) | (2) decimal $z = 2.11$ (Mm)

* Print variables with string

① Format specifier :

int $x = 123;$

Float $y = 5.8f.$

Console.WriteLine ("X = {0} \t Y = {1} ", x, y);

→ {0}, {1} →

ترتيب ال Variables
Index

If $x = \{0\}$ $y = \{23\}$

exception
→ runtime error
→ Caught by

Index (2) (3)

(2) ("X = " + x + "\t" + "Y = " + y)

(3)

③ Starting from (C# 6) $\Rightarrow$ (Interpolated string)

هيسر حلنا اننا زكيت انما ال variables جوا ال string
نفسه

بشرط اننا قبل اذ ال string

① انما ال \$ قبل البايه " _ " \$

→
writeline $\rightarrow$ Enter
write $\rightarrow$ xx

② انما ال variables مابيه { }

Console.WriteLine(\$ " x = {x} / y = {y} ")

Best way to ~~xxx~~ print

How To Read Input

→ Console.ReadLine(); $\Rightarrow$

بقرا الحابة ال داخل ال
انها string

فانت تسبقها بالانها string

* string str = Console.ReadLine();

هحتاج اني اخول بس # مفيش حاجة اسمها Casting ال string

← عليه (Parsing)

→ int x = int.Parse(str);
function static

30

* لو عليه ال Parsing تفت ما كن مفيش مشكلة

← لو عليه ال Parsing فشل ما كتب طابة مش int i; ولا

هيقول exception ← Input Isn't Correet

Swap Function

int a = 3, b = 6;
swap(a, b);

static void swap (int First, int second)

{
int Temp = First;
First = second;
second = Temp; }

معترض عليه من static
ال static مسيون غير static

∴ Console.WriteLine("# " A = {a} & B = {b} ");

طيب كده من الـ swap ليه

a, b → Integers

value type

(Calling By value)

عليه عليا

ليه SS

static void swap(
ref int First,
ref int second) {
}

لو عاوز نيه الموضوع ده يحصل؟؟

Call By Reference

(31)

→ على أنه مستقبل حاجة By Refrule

كيف نتكتب (Ref)

~~swap~~

→ static void swap (ref int first, ref int second) {

int Temp = first

~~int~~ first = second; second = Temp; }

int a = 3, b = 5
swap (ref a, ref b)

→ seeking for Readability.

→ example

static int divide (int num, int den)
{

return num / den;

① هيرج ناتج القسمة

② أنا عاوزه اترجع بامي البقية ..

How?

هنا انا عاوزه اترجع معلومتي

int num, int den,
ref int rem)

rem = num % den

By Ref

← هيرج

Int rem = 0;

int result = Divide(a, b, ref rem)

int rem;

int result = Divide(a, b, out rem);

Divide Function

طب في ال

out re

out

لو مستخدم

حل مستخدم اعل intillization لو ?? rem = rem % den

XX عس هيفيلد نطع م ال Function الا وانب - كامل
Intillization of the remaining

Parse Function

مسكنا اننا نبري Exception

Run time error

Try Parse

Parse Function

← كوفر نسق اخرى م ال

True or false

Boolean

To Tell me

(34) The parse Process succeed or failed

→ and If the Parsing Process went good, will put in the output value, send out not ref → without initialization.

int a;
if (int.TryParse(^{str}out a)) { Console.WriteLine (a) };
=

→ if (int.TryParse(str, out int a) { — }

→ another type to convert string to Integer
at class Convert

→ Convert.ToInt32(str);

→ another concept → dis card  → underscore